

Mohammad Askari

M.Sc. Student in Optimization Methods

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Research Summary

I am currently an M.Sc. student in Industrial Engineering (Optimization Methods) at the University of Tehran, with a strong academic and technical background in applied machine learning, optimization modeling, and data-driven decision systems. My primary research interest is in developing intelligent optimization frameworks that combine learning-based models with classical mathematical programming. I am actively looking for a supervisor engaged in research on optimization, machine learning, or decision intelligence.

Education

- 2025–Present **M.Sc. in Industrial Engineering (Optimization Methods)**, *Tehran University*, Tehran, Iran
Expected graduation: 2027
interests : Optimization, Machine Learning, Data-driven Decision Making.
- 2019–2025 **B.Sc. in Electrical Engineering (Control Systems)**, *University of Tehran*, Tehran, Iran
Relevant courses: Linear Algebra, Operations Research, Intelligent Systems, Introduction to Data Science, Linear Control, Modern Control, Digital Control.

Research Interests

Optimization Algorithms, Machine Learning, Data-driven Decision Making, Applied Artificial Intelligence.

Selected Projects

- B.Sc. Thesis **Markowitz Mean–Variance Portfolio Optimization Based on Machine Learning Forecasting** This project integrated deep learning models with classical financial optimization to enhance stock portfolio performance. A hybrid framework combining Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) networks was designed to forecast future stock prices of the SandP 500 index. The model employed robust statistics (Huber's M-estimator) for noise reduction and trained on historical data from 2013–2018 to predict 21-day returns. Experimental results demonstrated that the CNN–BiLSTM model outperformed conventional LSTM and BiLSTM architectures in forecasting accuracy, achieving lower MSE and SMAPE values. When integrated into the Markowitz mean–variance framework, the optimized portfolio improved its Sharpe ratio from approximately 1.1 (equal-weighted baseline) to about 1.6, confirming a substantial gain in return-to-risk efficiency achieved through the hybrid deep learning and optimization approach. The predicted returns were then integrated into the Markowitz mean–variance optimization model to construct an optimal portfolio. Portfolios formed from the top 10 predicted stocks consistently achieved higher Sharpe ratios compared to equally weighted portfolios with similar risk levels. The findings show that combining intelligent stock selection through deep learning with efficient weighting via Markowitz optimization can significantly increase return-to-risk efficiency, leading to more stable and profitable investment strategies. available on GitHub

- Course Project **Linear Algebra Project** Implemented a natural language processing (NLP) pipeline using linear algebra concepts such as vector spaces and matrix decomposition. Performed text preprocessing, tokenization, and frequency visualization (Top-30 words, word cloud). Constructed Bag-of-Words matrices and applied Singular Value Decomposition (SVD) for dimensionality reduction and reconstruction error analysis. Calculated cosine similarity and Euclidean distance between word vectors for semantic comparison. Applied SVM classification in the latent space to label unseen documents, achieving over 73 percent accuracy in multi-category text prediction. available on GitHub
- Course Project **Intelligent Systems Project – Deep Q-Learning for Breakout** Designed and implemented a Deep Q-Learning (DQL) agent to play the Breakout game autonomously. Built a deep neural network with target and prediction networks to stabilize training using Bellman updates and epsilon-greedy exploration. Tuned hyperparameters such as gamma, epsilon decay, and batch size, and analyzed convergence under constant and decaying epsilon policies. Discussed the role of convolutional layers for vision-based state representations, demonstrating a solid understanding of reinforcement learning mechanics and agent-environment interaction. available on GitHub
- Course Project **Operation Research Project** This project involved the end-to-end implementation of two core optimization problems using Python, covering transportation, numerical optimization, and route planning. **Transportation Optimization:** Formulated and solved a multi-source, multi-destination linear programming model to minimize total shipping costs between European cities using the Pyomo library and GLPK solver. Conducted sensitivity analysis on different supply–demand configurations to assess model stability and optimality. **Gradient Descent Implementation:** Derived the analytical gradient and optimal step-size formula for a quadratic cost function, then implemented and compared fixed-step and adaptive-step gradient descent algorithms to evaluate convergence behavior and computational efficiency. This project deepened my expertise in mathematical modeling, linear and nonlinear optimization, and algorithmic problem-solving with Python for real-world decision systems. available on GitHub
- Course Project **Real Time Payment Data Pipeline** This project involved building a full real-time payment data pipeline for a simulated financial service provider (“Darooghe”) using Apache Kafka, Apache Spark (PySpark), and MongoDB. I developed Kafka producers and consumers to stream and validate live payment transactions, detecting anomalies such as inconsistent amounts, clock drift, and device spoofing. Batch jobs built with PySpark performed commission efficiency analyses, customer segmentation, and temporal transaction pattern discovery. A Spark Streaming layer processed live Kafka data for real-time fraud detection based on velocity, geographical distance, and amount anomalies, while continuously generating commission and merchant performance metrics. Aggregated results were stored and queried through MongoDB for visualization dashboards created with Python libraries (Plotly/Matplotlib). The system demonstrated an end-to-end scalable architecture capable of fraud monitoring, dynamic pricing simulation, and data-driven business insights for fintech environments. available on GitHub
- Course Project **Machine Learning Challenges** This project focused on solving three real-world machine learning challenges like classification, regression, and recommendation as part of a competitive Kaggle-based data science assignment. In the classification task, I developed a predictive model for cancer patient survival using demographic, clinical, and treatment data, applying extensive preprocessing, feature encoding, and model evaluation with accuracy, precision, recall, and F1-score. The regression task involved predicting daily bike rental counts from meteorological and calendar features, using feature selection (p-values, correlations), multiple regression algorithms, and error metrics such as RMSE and R^2 . Finally, in the recommender system task, I designed a machine-learning-based movie recommendation engine that integrates user trust relationships and historical ratings to predict unseen user–movie ratings. The project demonstrated end-to-end proficiency in data preprocessing, model selection, evaluation, and competition-oriented machine learning workflows without deep learning models. available on GitHub

Personal Project **Smat Call Assistant Project** Smart Call Assistant is an AI-driven voice automation system that listens to incoming calls, transcribes speech using OpenAI Whisper, analyzes user intent and sentiment through natural language processing, generates intelligent responses via GPT-based models, and converts them back into speech using Google TTS. Built with Flask and SQLite, it features modular components for audio preprocessing, intent detection, and real-time voice interaction, enabling a fully autonomous conversational flow. The project demonstrates expertise in integrating large language models, speech technologies, and backend orchestration for developing scalable, human-like call-assistant systems. available on GitHub

Technical Skills

Programming Python, MATLAB, C
Optimization Pyomo, cvxpy, MATLAB Optimization Toolbox
Tools
Frameworks TensorFlow, Scikit-learn
Tools Git, Linux, LaTeX
Languages Persian (Native), English (Upper-Intermediate)

Online Profiles

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